

Computer Science - K-12 Level Assessment 25

Instructions

- Answer all questions.

Multiple Choice

Question 1

(2 marks)

Which of the following has the greatest potential for compromising a user's personal privacy?

- A. A group of cookies stored by the user's Web browser
- B. The Internet Protocol (IP) address of the user's computer
- C. The user's e-mail address
- D. The user's public key used for encryption

Question 2

(2 marks)

Let `l = [1,2,3,4]`. What is `max(l)` in Python 3?

- A. 1
- B. 2
- C. 3
- D. 4

Question 3

(2 marks)

In Python 3, let `r = lambda q: q * 2`. What is `r(3)`?

- A. 2
- B. 6
- C. 3

D. 1

Question 4

(2 marks)

In python 3, which of the following is floor division?

A. /

B. //

C. %

D. |

Question 5

(2 marks)

Which is the smallest asymptotically?

A. $O(1)$

B. $O(n)$

C. $O(n^2)$

D. $O(\log n)$

True / False

Question 1

(2 marks)

In Python 3, the expression ['Hi!'] * 4 produces the list ['Hi!', 'Hi!', 'Hi!', 'Hi!'].

True

False

Question 2

(2 marks)

In Python 3, accessing the first element of the tuple ('abcd', 786, 2.23, 'john', 70.2) returns 'abcd'.

True

False

Question 3

(2 marks)

The boolean expression $a[i] == \text{max} \parallel \neg(\text{max} != a[i])$ simplifies to $a[i] != \text{max}$.

True

False

Question 4

(2 marks)

The program may have bugs even if extensive testing found no errors.

True

False

Question 5

(2 marks)

Python variable names are not case-sensitive and can be used interchangeably regardless of capitalization.

True

False

Long Form

Question 1

(10 marks)

Write a function to find the first duplicate element in a given array of integers.

Type your answer here...



Question 2

(10 marks)

Write a function to find three closest elements from three sorted arrays.

Type your answer here...



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